

CHAPTER 3

SYSTEM DESIGN

3.1 Introduction

System design defines the architecture, modules, and interactions of the EduGenie Learning Assistant. It converts the requirements gathered during analysis into a structured implementation plan. The design emphasizes modularity, scalability, maintainability, and ease of integration with AI services.

3.2 Overall Architecture

EduGenie follows a three-tier architecture consisting of the presentation layer, application layer, and AI service layer. Users interact through an HTML/CSS interface. Requests are processed by the FastAPI backend, which communicates with Google Gemini and LaMini-Flan-T5 to generate educational responses.

3.3 Modules

The application contains five major modules: Question Answering, Explanation, Quiz Generation, Summary Generation, and Learning Path Recommendation. Each module performs a dedicated educational task while sharing a common backend framework.

3.4 Database and Data Flow

The project uses lightweight JSON-based storage to maintain application data. User requests flow from the frontend to FastAPI, are processed by the appropriate AI model, and the generated output is returned to the browser. The modular data flow reduces complexity and improves response handling.

3.5 ER Diagram

The ER model contains entities such as User, User_Query, AI_Response, Quiz, Summary, and Learning_Path. Primary and foreign keys define relationships, ensuring organized storage of educational interactions and generated content.

3.6 System Workflow

The workflow begins when the user submits a request through the web interface. FastAPI validates the request, invokes the required AI module, processes the output, and displays the generated answer, explanation, quiz, summary, or learning path. Error handling ensures stable execution.

3.7 Design Advantages

The modular architecture supports code reusability, easy debugging, future enhancements, and integration of additional AI models. FastAPI provides high performance, while HTML templates ensure a simple user experience.

3.8 Chapter Summary

This chapter explained the architecture, modules, workflow, ER design, and overall system organization of EduGenie. These design decisions provide a strong foundation for implementation and future expansion.